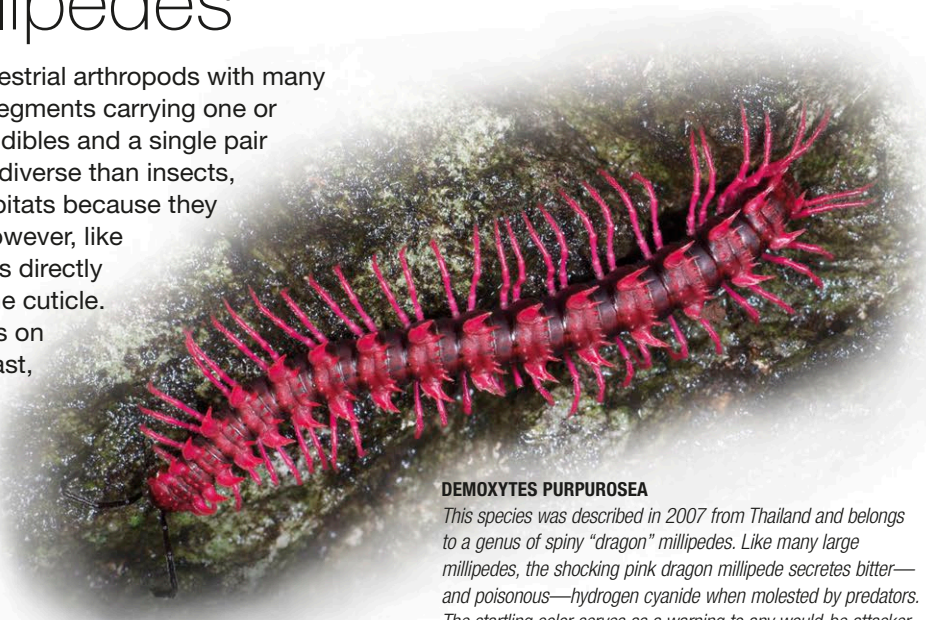


Centipedes and millipedes

PHYLUM	Arthropoda
SUBPHYLUM	Mandibulata
SUPERCLASS	Myriapoda
CLASSES	Chilopoda Diplopoda
ORDERS	21
FAMILIES	171
SPECIES	About 13,150

Centipedes and millipedes are terrestrial arthropods with many legs, each of their numerous body segments carrying one or 2 pairs. Their heads have biting mandibles and a single pair of antennae. Far less abundant and diverse than insects, they are confined to humid microhabitats because they do not have a waterproof cuticle. However, like insects, they breathe by taking gases directly into the body through openings in the cuticle. Centipedes have a single pair of legs on each trunk segment. They can run fast,

although some soil-living species move slower. They are carnivorous and have venomous claws to kill prey. Millipedes are all slow moving, with elongated, cylindrical bodies. Most of their trunk segments are fused together in pairs, called diplosegments, each bearing 2 pairs of legs. They are mostly herbivores or scavengers, using their toothed mandibles to chew rotting organic matter, soil, plants, algae, and moss.



DEMOXYTES PURPURESEA
This species was described in 2007 from Thailand and belongs to a genus of spiny “dragon” millipedes. Like many large millipedes, the shocking pink dragon millipede secretes bitter—and poisonous—hydrogen cyanide when molested by predators. The startling color serves as a warning to any would-be attacker.

Class Chilopoda

CENTIPEDES

Elongated and flattened, centipedes have a trunk made up of at least 16 segments, most of which carry one pair of legs; the last pair of walking legs is always longer than the rest. Most species are yellowish or brownish, and all have fine, sensory hairs on the body. There are 5 orders, 24 families, and 3,149 species.

Family Lithobiidae

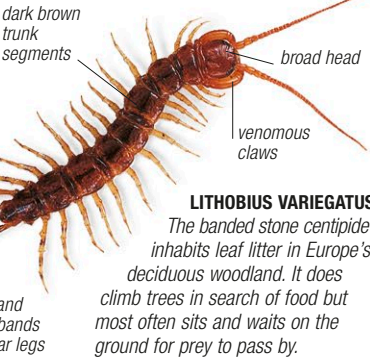
Lithobiids

Occurrence 975 spp. worldwide, especially in temperate regions of the Northern Hemisphere; in cracks, crevices



Lithobiids are mostly reddish brown but some can be brightly colored. The tough and flattened body is ¼–1¾ in (0.6–4.5 cm) long. There are 15 pairs of legs, and threadlike antennae with 75–80 segments. The young emerge from the egg

with 7 pairs of legs and add a leg-bearing trunk segment at each molt. Lithobiids have a lifespan of 5 to 6 years.



LITHOBIUS VARIEGATUS
The banded stone centipede inhabits leaf litter in Europe's deciduous woodland. It does climb trees in search of food but most often sits and waits on the ground for prey to pass by.

Family Scolopendridae

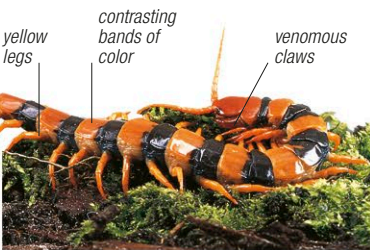
Scolopendrids

Occurrence 400 spp. worldwide, especially in subtropical and tropical regions; in soil, leaf litter, cracks, crevices



The largest centipede in the world, *Scolopendra gigantea* from South America, belongs to this family. The body length of scolopendrids ranges from 1¼–12 in (3–30 cm). Nocturnal hunters, they are robust enough to catch and kill mice and frogs with their venomous claws, the smaller species being the most deadly. Scolopendrids are typically brightly colored and may be yellow, red, orange, or green, often with dark

stripes or bands. They have 21 or 23 pairs of legs, and threadlike antennae with fewer than 35 segments. Usually, there are 4 ocelli, or simple eyes, on either side of the head. Females lay their eggs under soil, rocks, or loose bark.



SCOLOPENDRA HARDWICKEI
Tiger centipedes owe their common name to the markings on their body and their predatory habits. They are found in Southeast Asia, living under stones, rotting wood, and loose bark.

Class Diplopoda

MILLIPEDE

Despite their name, millipedes never have a thousand legs. Some have as many as 750, but even these species are slow runners. To defend themselves, most millipedes rely primarily on their tough exoskeletons, which are reinforced by calcium carbonate. When touched, many also coil up or produce toxic chemicals. This class has 9,973 species, classified in 16 orders and 147 families.

Family Julidae

Cylinder millipedes

Occurrence 900 spp. in the Northern Hemisphere; in soil, leaf litter, caves, under stones and rotting wood



These millipedes have rounded bodies that can range from ⅝–3¼ in (0.8–8 cm) in length. Most are dull-colored but a few have red, cream, or brown spots. The antennae are fairly long and slender. As in all millipedes, the females lay their eggs in nests in soil. There are typically 7 nymphal stages. Cylinder millipedes are common in Europe and Asia.



TACHYPODOIULUS NIGER
This species is commonly known as the black snake millipede. It has a black trunk and white legs, and is up to 2 in (4.8 cm) long. It inhabits woodland in Western Europe and climbs walls, trees, or shrubs to feed after dark.

Family Glomeridae

Pill millipedes

Occurrence c.200 spp. in warm and cool temperate regions of the Northern Hemisphere; in soil, caves



Measuring ¼–¾ in (0.2–2 cm) in length, these broad-bodied millipedes are either small and drably colored or large and brightly marked. The trunk is made up of 13 segments, and the shape of the dorsal plates allows the millipede to roll into a tight ball if threatened. Adults have 15 pairs of legs. Eggs are laid in soil, and when the young hatch, they initially have only 3 pairs of legs. Some members of this family, such as *Glomeris* species, may live up to 7 years.



GLOMERIS MARGINATA
The white-rimmed pill millipede can be found in soil and leaf litter in Europe, and parts of Asia and North Africa. When completely rolled up, it can be confused with the pill woodlouse (see p.584), although it has a shinier body.